LOCALLY CONSTRUCTIBLE SUBSETS OF THE TOPOLOGICAL SPACE CHARACTERIZED ON ETALE COVERS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file spaces-properties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 13 S10-geom published cards for spaces-properties.tex (prop-occupied.json, 2026-10-14); those 13 labels are excluded from this batch by design. For the 77 free labels: 0 occupied. The registry mentions field (7557 files) and focused-seeds (645) both point to the same S10-geom cards (dual-source check), no other occupation.

Source reference. Original reference 054C: sources/stacks-chow/tags.txt, source lines 6635–6635. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 054I: sources/stacks-chow/tags.txt, source lines 6641–6641. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 03H4: sources/stacks-chow/tags.txt, source lines 4553–4553. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 054K: sources/stacks-chow/tags.txt, source lines 6643–6643. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let S be a scheme. Let X be an algebraic space over S. Let $E \subset |X|$ be a subset. The following are equivalent*

1. *for every étale morphism $U \to X$ where U is a scheme the inverse image of E in U is a locally constructible subset of U,*
2. *for every étale morphism $U \to X$ where U is an affine scheme the inverse image of E in U is a constructible subset of U,*
3. *for some surjective étale morphism $U \to X$ where U is a scheme the inverse image of E in U is a locally constructible subset of U.*

Proof. By Properties, Lemma 054C we see that (1) and (2) are equivalent. It is immediate that (1) implies (3). Thus we assume we have a surjective étale morphism $\varphi : U \to X$ where U is a scheme such that $\varphi^{-1}(E)$ is locally constructible. Let $\varphi' : U' \to X$ be another étale morphism where U' is a scheme. Then we have

$$E'' = \mathrm{pr}_1^{-1}(\varphi^{-1}(E)) = \mathrm{pr}_2^{-1}((\varphi')^{-1}(E))$$

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where $\mathrm{pr}_1 : U \times_X U' \to U$ and $\mathrm{pr}_2 : U \times_X U' \to U'$ are the projections. By Morphisms, Lemma 054I we see that E'' is locally constructible in $U \times_X U'$. Let $W' \subset U'$ be an affine open. Since pr_2 is étale and hence open, we can choose a quasi-compact open $W'' \subset U \times_X U'$ with $\mathrm{pr}_2(W'') = W'$. Then $\mathrm{pr}_2|_{W''} : W'' \to W'$ is quasi-compact. We have $W' \cap (\varphi')^{-1}(E) = \mathrm{pr}_2(E'' \cap W'')$ as φ is surjective, see Lemma 03H4. Thus $W' \cap (\varphi')^{-1}(E) = \mathrm{pr}_2(E'' \cap W'')$ is locally constructible by Morphisms, Theorem 054K as desired. $\square$